

Michael Avendano

Software Engineer II

Honolulu, HI (open to relocation) | mavend2@gmail.com
linkedin.com/in/michael-avendano | github.com/avenmia | michaelavendano.com

SUMMARY

Full-stack software engineer with seven years of experience in .NET, TypeScript, and Azure. Builds and modernizes production APIs, data layers, and serverless services. Enjoys solving problems and making systems measurably better through the use of CI quality gates, automated test coverage, and documentation that shortens ramp-up.

PROFESSIONAL EXPERIENCE

Pangaea Information Technologies — Software Engineer II

Jan 2019 – Present

TMPT — Training Support Systems-Enterprise Master Planning Tool — Enterprise planning system for U.S. Army range and training management, supporting 350+ active globally distributed users.

- Led the platform migration from .NET Framework 4.8 to .NET 10, including Entity Framework 6 to EF Core; accelerated conversion with Claude Code while gating every change behind unit tests and user acceptance testing.
- Designed, built, and maintained OData APIs over Entity Framework and SQL Server, enforcing policy-based access control (PBAC) across Army-wide project data.
- Rewrote the developer documentation to capture Army range-planning domain knowledge new engineers had only picked up by asking, cutting time to a first contribution.

CEC Taking Stock Online — Public data platform unifying pollutant records from ~40,000 North American industrial facilities into an interactive query, mapping, and reporting tool.

- Led the migration off NancyFX to ASP.NET MVC, rewriting bootstrapping, routing, and exception filtering as three reviewable phases using AI-assisted tooling with unit-test gates.
- Audited landing-page performance with Chrome DevTools traces and implemented fixes cutting image payload 60% (4.7 MB → 1.9 MB) and Largest Contentful Paint 58% (17.8s → 7.4s).
- Audited the site with Google Lighthouse for ADA/WCAG accessibility issues and implemented the fixes.

FlowKaizen Performance Tools — Multi-tenant SaaS platform visualizing patient and workflow through hospital departments with EMR integration.

- Built the platform's serverless REST API in TypeScript on Azure Functions backed by Cosmos DB, supporting multi-tenant data access for hospital clients; tuned indexes across all containers so common queries stayed fast as data grew.
- Defined the platform's Azure infrastructure as code in Terraform, replacing manual portal provisioning with reproducible, version-controlled environments.

VOLUNTEER EXPERIENCE

Code With Aloha

Jan 2022 – Present

Lead Organizer (Jan 2026 – Present) · Volunteer Developer (Jan 2022 – Jan 2026)

- Promoted to Lead Organizer after four years as a Volunteer Developer; direct 15–20 volunteers.

HIERR Survey Tool — Civic engagement platform for the Hawai'i Office of Planning & Sustainable Development (OPSD).

- Led design and deployment of the survey tool, directed a volunteer team of 5–7 and partnered with state agency stakeholders to align the platform with OPSD goals.
- Developed a wrapper for the Polis survey to collect demographic data for the state. Utilized Next.js, React, TypeScript, Prisma, and Tailwind; deployed across 20+ community workshops statewide.

Toastmasters International, Nu'uanu Club

June 2024 – Present

Club President (June 2026 – Present) · Member (June 2024 – June 2026)

- Elected Club President after two years as a Member; lead meetings, officer coordination, and membership growth.

EDUCATION

University of Illinois at Chicago — B.Sc., Computer Science (Dec 2018) | B.Sc., Physics (May 2015)

CERTIFICATIONS

Certifications: CompTIA Security+ (SY0-601) | Microsoft Certified: Azure Developer Associate

Awards: Hawaii Annual Code Challenge — 1st Place 2024 (Aloha Birds) | 4th Place 2023 (Marine Debris)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Python, SQL

Frameworks & Libraries: .NET 10, .NET Framework 4.8, ASP.NET MVC, Entity Framework Core, OData, React 19, Next.js, Node.js, Prisma, Tailwind

Cloud & Data: Azure, Azure Functions, Cosmos DB, SQL Server, Terraform, OAuth / JWT, OpenTelemetry, OpenAPI

Testing & Tooling: NUnit, Jest, Playwright, GitHub Actions, ESLint / Prettier, Vite, Git, Linux, Vim, IIS, Windows